

Aswin M Kumar

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SUMMARY

Electrical and Electronics Engineering student building AI-powered systems at the intersection of embedded hardware, IoT, and applied machine learning. Delivered production-deployed LLM pipelines, won recognition at national hackathons and industry expos, and completed government-funded aerospace training. Targeting roles in embedded systems, IoT engineering, and AIoT product development.

EDUCATION

College of Engineering Trivandrum (CET) Kerala, India
B.Tech – Electrical and Electronics Engineering | CGPA: 8.21/10 2023 – Present (Graduating May 2027)
• Relevant Coursework: Embedded Systems, Control Systems, Signal Processing, Machine Learning, IoT Applications

TECHNICAL SKILLS

Programming: Python, C, JavaScript, Flutter/Dart
Frameworks & Tools: Flask, OpenCV, Arduino, ESP32, Arduino Cloud, KiCAD, MATLAB/Simulink, ANSYS, Git
Domains: Embedded Systems, IoT Systems, Machine Learning, Computer Vision, LLM Integration, PCB Design
Platforms: GitHub Actions, Render, GitHub Pages, Wokwi

EXPERIENCE

IoT Intern June 2025
Keltron (Kerala State Electronics Development Corporation) Thiruvananthapuram, Kerala
• Prototyped IoT systems using ESP32 and Arduino Uno interfaced with ultrasonic, thermal, and LiDAR sensors
• Deployed cloud-connected monitoring systems on Arduino Cloud; implemented relay-based automation and real-time device state tracking
• Built functional prototypes (water level monitor, temperature array, traffic signal controller) and authored technical report on IoT device architecture

Flutter Intern Dec 2025 – Apr 2026
PgBee, TBI CET Thiruvananthapuram, Kerala
• Implemented 7 core screens for a PG/hostel discovery app from Figma designs using Flutter in a startup environment
• Delivered responsive UI within Git-based agile workflows; app currently in beta on Play Store

PR Coordinator 2025 – Present
Internship Cell, CET Thiruvananthapuram, Kerala
• Designed and deployed InternAgent and InternBuddy – AI tools that cut internship post creation time from 45 minutes to under 20 minutes
• Coordinated student-recruiter communication and managed outreach campaigns across channels

Participant – Drone/UAS Technologies Bootcamp June 2025
MeitY / SwaYam Programme, NIT Calicut Calicut, Kerala
• Completed 40-hour government-funded programme covering aerodynamics, drone assembly, flight operations, and CFD simulation
• Executed ANSYS Mechanical and Fluent structural and CFD analyses on airfoil designs; produced simulation report
• Designed, fabricated, and flew a straight-wing glider; secured **3rd place** among 28+ teams in endurance flight competition

Membership Coordinator & Event Organiser 2025 – Present
SAEINDIA – College of Engineering Trivandrum Thiruvananthapuram, Kerala
• Organised CADClash (10+ participants including first inter-college entry from AIT Pune) and Biomimicry ideathon (12 participants) at Tech Rave, SAEINDIA CET Technical Fest
• Managed SAEINDIA memberships and compliance documentation for BAJA SAEINDIA 2026

PROJECTS

InternAgent – PR Content Automation | *Flask, Python, LLaMA 3.1 (NVIDIA NIM), Render* [Live](#)
• Cut internship poster creation time from 45 minutes to 10–20 minutes via a production-deployed LLM pipeline on Render
• Built SSRF-safe scraping pipeline with Cloudscraper fallback; structured text extraction handling 20k+ characters per document
• Integrated LLaMA 3.1 (70B) via NVIDIA NIM with prompt-engineered workflows; dynamic frontend with QR-based application linking

InternBuddy – Student AI Platform | *Flask, Python, LLaMA 3.1 (NVIDIA NIM), BeautifulSoup* [Live](#)
• Full-stack AI platform for CET Internship Cell: skill gap detection, eligibility validation, and resume-to-role matching
• Engineered LLM-backed APIs for structured extraction from URLs and documents; production-deployed with CSRF protection, rate limiting, and anti-bot handling

LithiumSense – BMS Architecture | *System Design, Fault Analysis, Technical Presentation*

- Architected a fire-hardened Battery Management System with a 5-stage emergency response pipeline targeting sub-500ms response across overvoltage, undervoltage, overcurrent, short-circuit, and thermal runaway faults
- Designed a 900°C-rated survivable black-box data module for post-incident forensic analysis
- **Trophy winner** at Allianz Risk Consulting Compass-360 Expo; presented to industry risk and engineering professionals

Trash Nada – AI Waste Management | *Python, React, OpenCV, TensorFlow, IoT*

[GitHub](#)

- Built frontend and IoT integration for an AI-based waste classification system (~95% model accuracy) with real-time hotspot mapping
- **Top 10 / 133 teams** at Satwa25 National Hackathon (36-hour competition, 2025)

Gesture-Controlled Interface | *Arduino Leonardo, MPU6050, I2C, C*

[GitHub](#)

- IMU-based motion control interface achieving ~50ms end-to-end latency for real-time gesture input

NOT_KTU Results | *HTML5, CSS3, JavaScript, Tailwind CSS, Tone.js*

[GitHub](#)

- Satirical university portal parody with a progressive multi-rule password game and real-time audio feedback; **selected for TinkerHub Useless Projects 2.0**

Resume CI/CD Pipeline | *GitHub Actions, LaTeX, Git*

- Automated LaTeX resume compilation and PDF deployment on every commit; version-controlled resume with continuous deployment

ACHIEVEMENTS

Top 10 / 133 teams – Satwa25 National Hackathon, Saintgits College of Engineering (2025)

Trophy Winner – Allianz Risk Consulting Compass-360 Expo, LithiumSense BMS Architecture (2025)

3rd Place – Endurance flight competition, MeitY/SwaYaan Drone/UAS Bootcamp, NIT Calicut (2025)

CERTIFICATIONS

Google AI Essentials – Google | Cybersecurity Essentials – Cisco | Industrial Automation & PLC Systems – Bosch | PCB Design Workshop – IET CET